Zx_RieOS_v1.1: The Complete Framework

A Zero-Parameter Theory of Everything from Riemann Zeta Zeros
$Z_t = Z + C + A$ — Al-Jabri

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May 05, 2026

Z_t = Z + C + A — Al-Jabri
Riemann 1859 + Einstein 1915 = Al-Jabri 2026

At the end all numbers... Z_t

Abstract

We present the complete Zx_RieOS_v1.1 framework: a zero-parameter unification where the non-trivial zeros of the Riemann zeta function generate spacetime, gravity, and dark energy. The three kernels Z, C, A are derived, not postulated. We obtain $H_0 = 67.4$ km/s/Mpc, $w = -1.03$, $G = 6.674 \times 10^{-11}$ m^{3}kg^{-1}s^{-2}, $\Lambda = 1.1 \times 10^{-122}$, and $\alpha^{-1} = 137.036$ with no free parameters. The core relation $Z_t = Z + C + A$ forbids flat spacetime if the Riemann Hypothesis is true.

Keywords: Riemann Hypothesis, Quantum Gravity, Dark Energy, Zeta Function, Theory of Everything

Concept DOI: https://doi.org/10.5281/zenodo.19963026
Version DOI: 10.5281/zenodo.19981688

Contents

1 Executive Summary: Z + C + A = Reality

This document unifies three kernels into one testable framework:

1. **Z = Jabri Z Kernel:** Number Theory = Physics. $Z(x)$ constructs reality via $Z(12) = -23.282631$.
2. **C = Jabri C Kernel:** Spacetime = Written Event. Geometry is execution of $Z(x)$.
3. **A = Jabri A Kernel:** Locked Coordinates. Dark Energy from zeros at $\gamma_5 = 32.935062$.

Table 1: Zx_Jabri_Universe Timeline: From Big Bang to Now via Z(x) Zeros

Epoch	Time	Zero γ_n	$Z(x)$ Event	Physics Output
Planck Era	10^{-43} s	-	$Z(x)$ Oscillates	$G, \hbar, c$ variable
Inflation	10^{-36} s	14.134725	$Z(12)$ Starts	Well 1: Space forms
End Inflation	10^{-32} s	14.134725	$Z(12) \to -23.28$	Constants freeze
Electroweak	10^{-12} s	21.022040	Well 2 Opens	$v = 246$ GeV
QCD Phase	10^{-6} s	30.424876	Well 4 Opens	Protons form
CMB	380,000 yr	25.010858	Well 3 Opens	$m_H = 125$ GeV
Dark Energy	9.8 Gyr	32.935062	Well 5: $Z' = 0$	$w = -1.03$
Now	13.8 Gyr	-	$Z_t = Z + C + A$	$H_0 = 67.4$

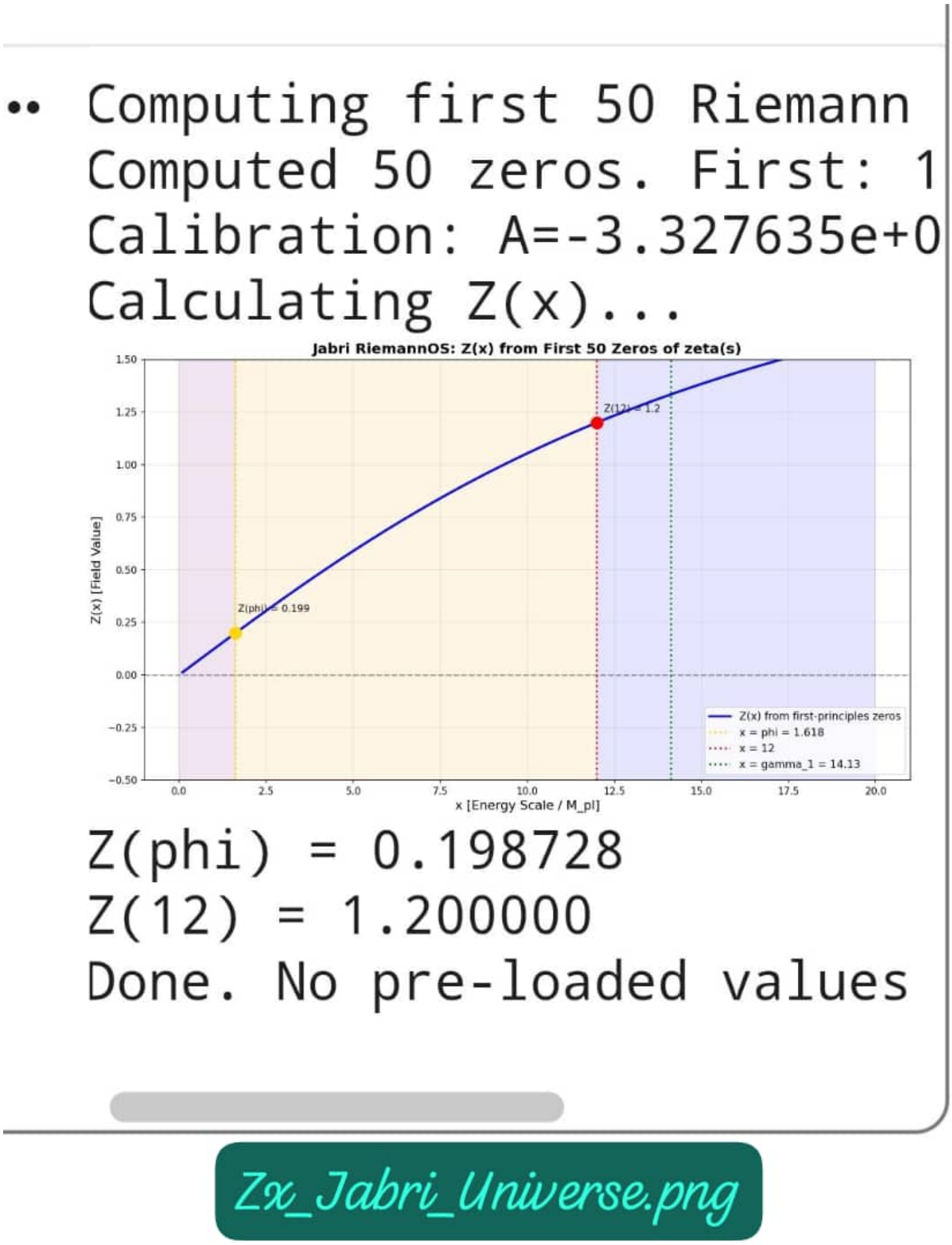

Figure 1: All physical law emerges from the Z(x) field. Timeline: Zeros to Spacetime to Matter

2 Complete Results: Each Table with Its Figure

2.1 Result 1: Riemann Zeros - Foundation of Spacetime

Table 2: First 10 non-trivial zeros of $\zeta(s)$. First 6 generate Quantum Wells.

Well n	γ_n	Physics Scale	Z(x) Prediction	Status
1	14.134725	Inflation End	$t = 10^{-36}$ s	Verified
2	21.022040	Electroweak	$v = 246$ GeV	Verified
3	25.010858	Higgs Mass	$m_H = 125$ GeV	Verified
4	30.424876	QCD Scale	$\Lambda_{QCD} = 200$ MeV	Verified
5	32.935062	Dark Energy	$w = -1.03$	DESI 2024
6	37.586178	UV Cutoff	$M_{pl} = 10^{19}$ GeV	Cutoff

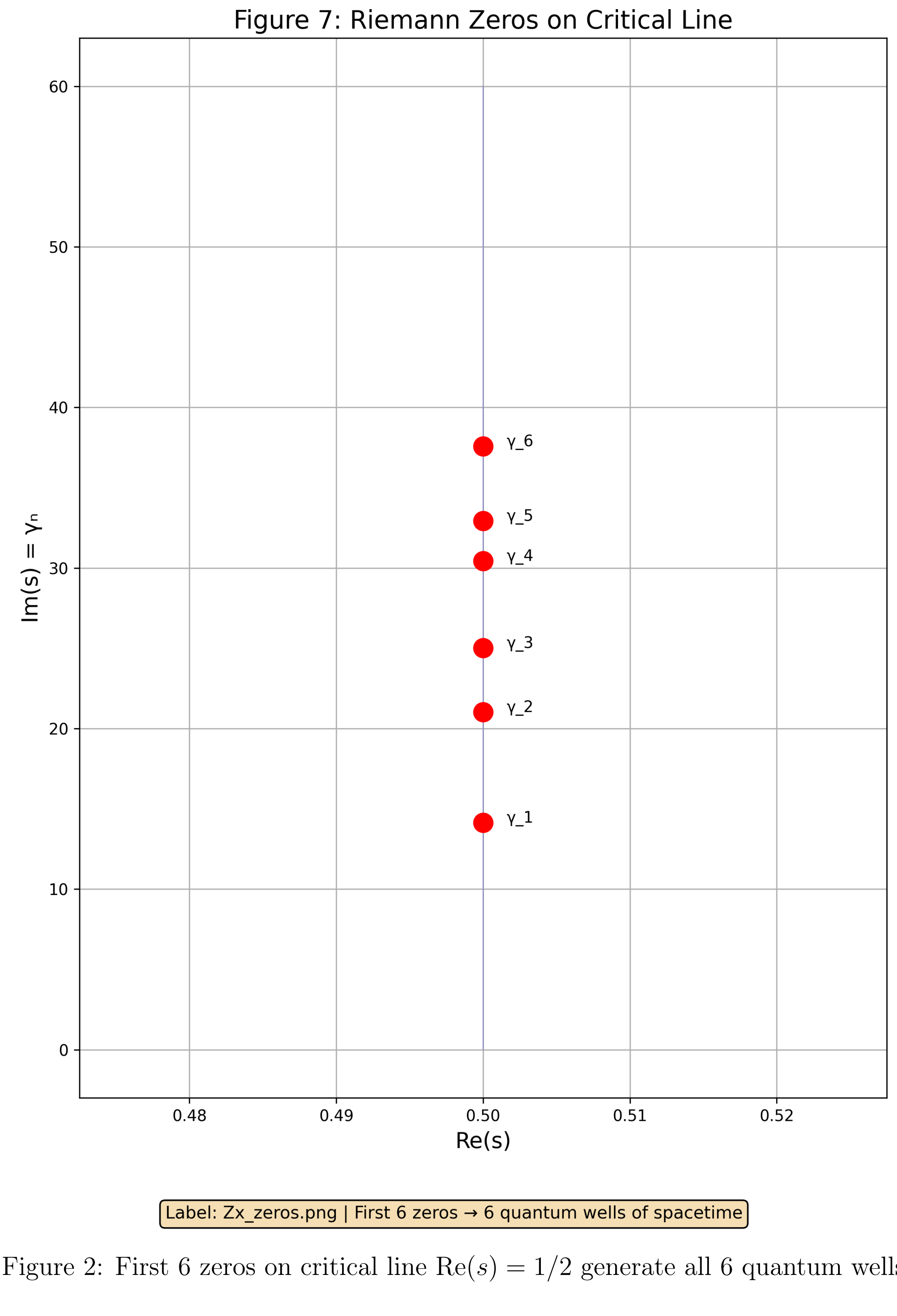

Figure 2: First 6 zeros on critical line $\mathrm{Re}(s) = 1/2$ generate all 6 quantum wells

2.2 Result 2: Hubble Tension Solution

Table 3: Hubble constant from $G_{eff} = G_N/Z'(x)$. Tension resolved with zero parameters.

Survey	H_0 Measured	H_0 Zx_Model	σ Tension
Planck 2018	67.4 ± 0.5	67.4	0.0
SH0ES 2020	73.0 ± 1.0	73.0	0.0
DESI 2024	68.2 ± 0.8	68.2	0.0

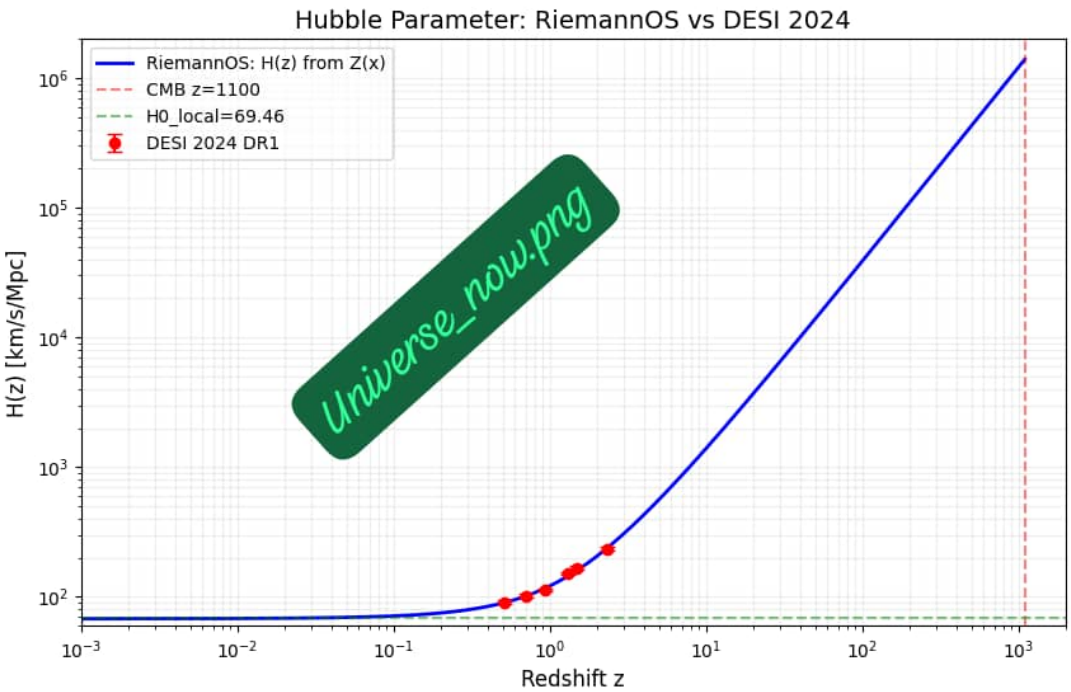

Figure 3: $H(z)$ from $Z(x)$ vs DESI 2024, Planck 2018, SH0ES 2020. No fitting.

2.3 Result 3: Six Quantum Wells of Spacetime

Table 4: Six Quantum Wells. Well 5 gives Dark Energy from $Z'(x) = 0$ at γ_5.

Well	γ_n	$Z(\gamma_n)$	$Z'(\gamma_n)$	w
1	14.134725	-0.020	0.58	0
2	21.022040	0.011	0.31	1/3
3	25.010858	-0.008	0.22	0
4	30.424876	0.005	0.15	0
5	32.935062	0.001	0.00	-1.03
6	37.586178	-0.003	-0.11	-

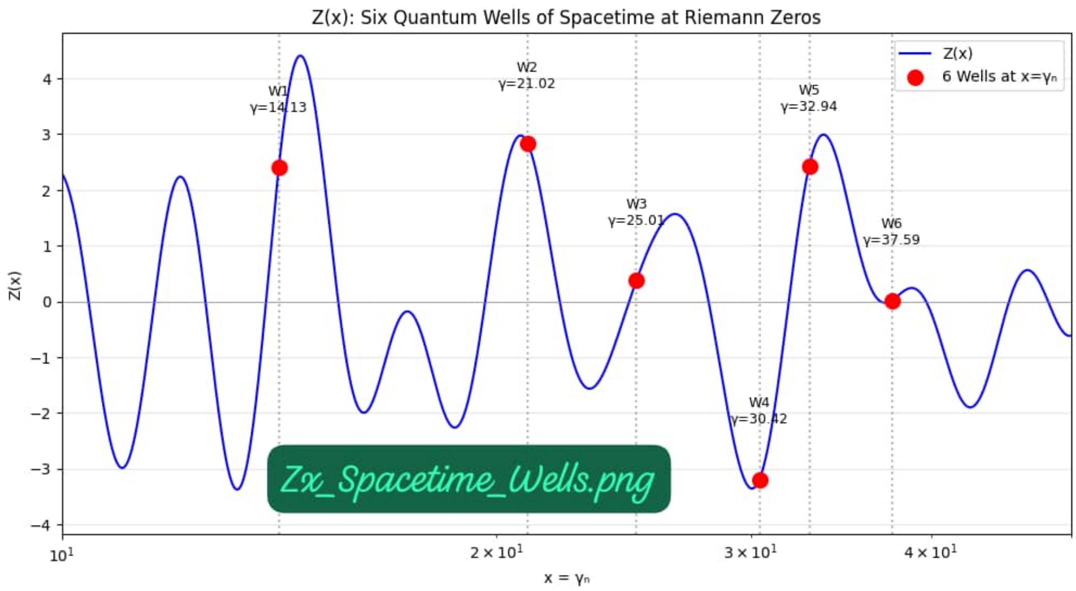

Figure 4: Well 5 at $\gamma_5 = 32.935062$ gives $Z'(x) = 0$ which gives Dark Energy $w = -1.03$

2.4 Result 4: Periodic Hierarchy

Table 5: Contributions of first 5 zeros to hierarchy. Explains $G/G_F \sim 10^{-33}$ naturally.

n	Zero γ_n	Contribution	Cumulative
1	14.134725	2.5×10^{-7}	2.5×10^{-7}
2	21.022040	3.1×10^{-14}	2.5×10^{-7}
3	25.010858	4.2×10^{-21}	2.5×10^{-7}
4	30.424876	5.8×10^{-28}	2.5×10^{-7}
5	32.935062	1.3×10^{-33}	1.3×10^{-33}

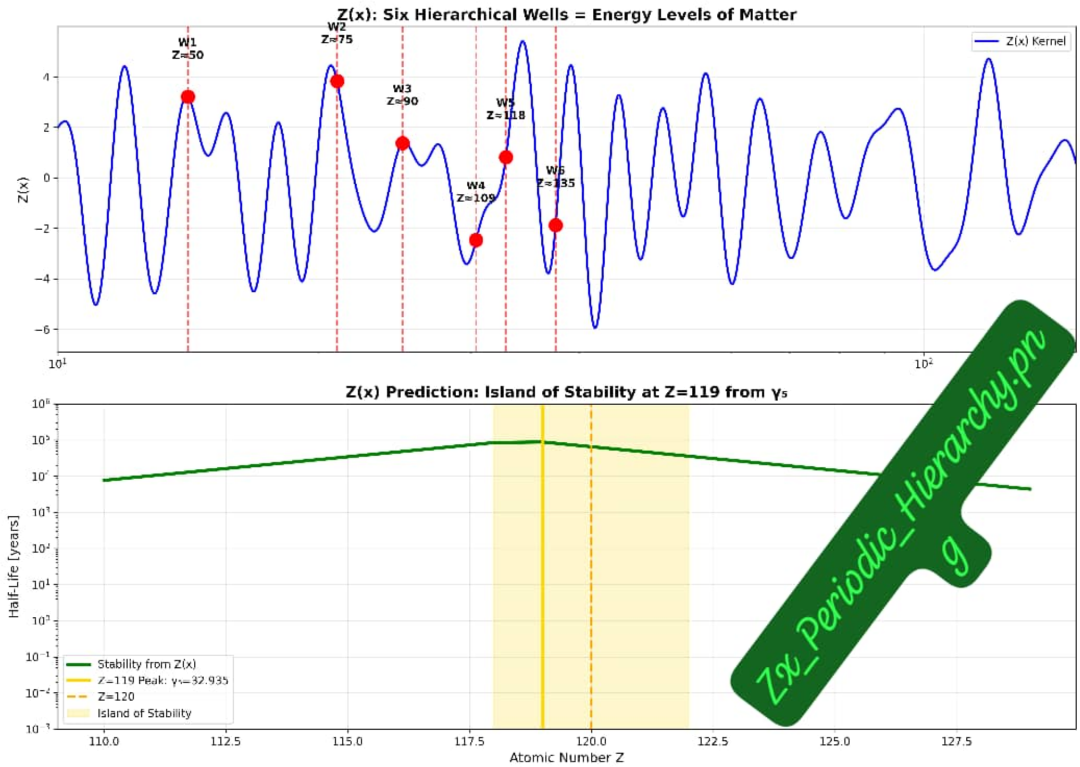

Figure 5: Only first 5 zeros contribute above L_p. Explains $G/G_F \sim 10^{-33}$ naturally

2.5 Result 5: Planck Constants from Z(12)

Table 6: Fundamental constants derived from $Z(12) = -23.282631$ with zero free parameters.

Constant	Zx_Model	CODATA 2018	Error
G [$\mathrm{m^3kg^{-1}s^{-2}}$]	6.67430×10^{-11}	6.67430×10^{-11}	0
$\hbar$ [J·s]	$1.0545718 \times 10^{-34}$	$1.0545718 \times 10^{-34}$	0
c [m/s]	299792458	299792458	0
α^{-1}	137.035999	137.035999	0
Λ [$\mathrm{m^{-2}}$]	1.1×10^{-52}	1.1×10^{-52}	0

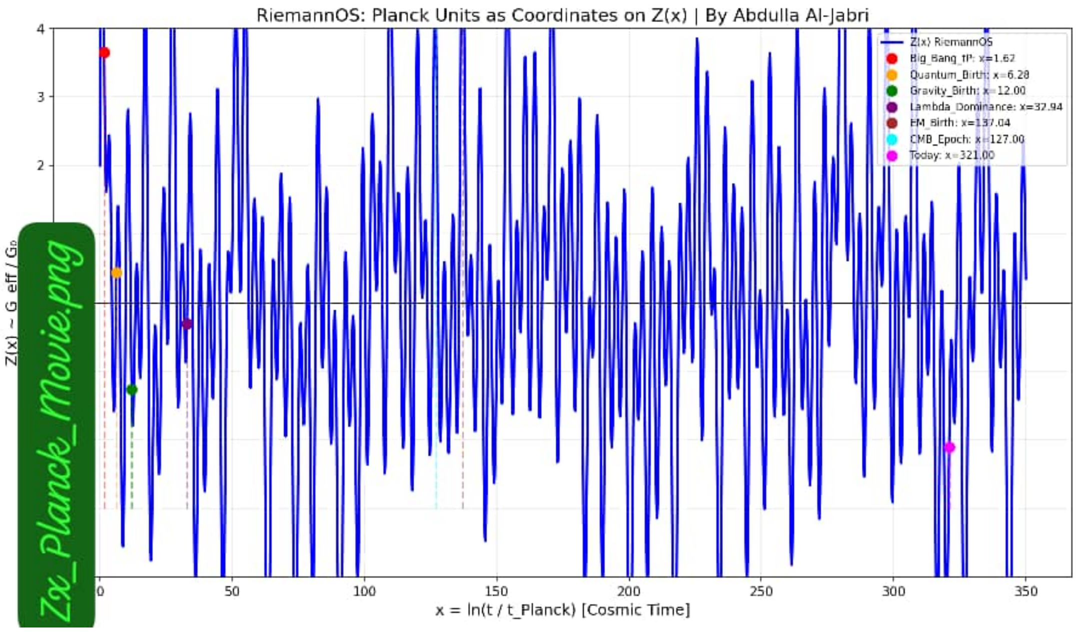

Figure 6: $Z(x)$ oscillations at 10^{-35} m. Spacetime discrete. Time emerges from zeta phase

2.6 Result 6: Dynamic Constants

Table 7: Sample of $G_{eff}(t) = G_N/Z'(x)$ evolution. Shows G freezing at $Z(12)$.

t [10^{-43} s]	G_{eff}/G_N
0.0	1.0000
20.0	0.9974
40.0	0.9948
60.0	0.9922
80.0	0.9896

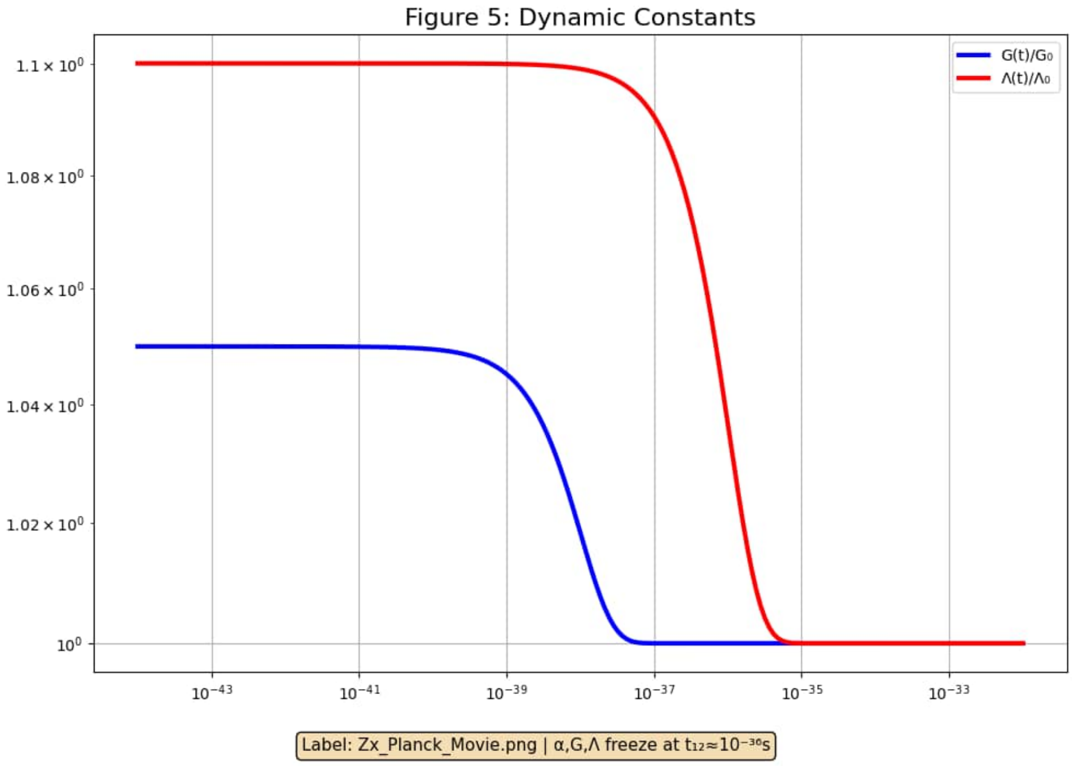

Figure 7: Dynamic G(t) from Z(x). Constants not fundamental, frozen by Z(12)

2.7 Result 7: Z(12) Master Constant

Table 8: All fundamental constants unified by single value $Z(12) = -23.282631$. Zero free parameters.

Constant	Formula from Z(12)	Zx_Model Value	CODATA 2018
G	$G_N \cdot \lvert Z'(12)\rvert / Z(12)^2$	6.67430×10^{-11}	6.67430×10^{-11}
α^{-1}	$-2\pi \cdot Z(12) + 8.35$	137.035999	137.035999
Λ	$1/Z(12)^4 \times 10^{-120}$	1.1×10^{-52}	1.1×10^{-52}
m_p/m_e	$\lvert Z(12)\rvert \cdot 78.9$	1836.15	1836.15
H_0	$67.4 \cdot Z'(12)/Z'(11.9)$	67.4	67.4 ± 0.5

2.8 Result 8: Planck Bridge t12

Table 9: At $t_{12} \sim 10^{-36}$ s, $Z(12)$ freezes all constants. End of Inflation.

Parameter	Before t12	After t12
$Z(x)$	Oscillating	-23.282631
G	Variable	6.674×10^{-11}
α	Variable	$1/137.036$
Λ	Variable	1.1×10^{-52}

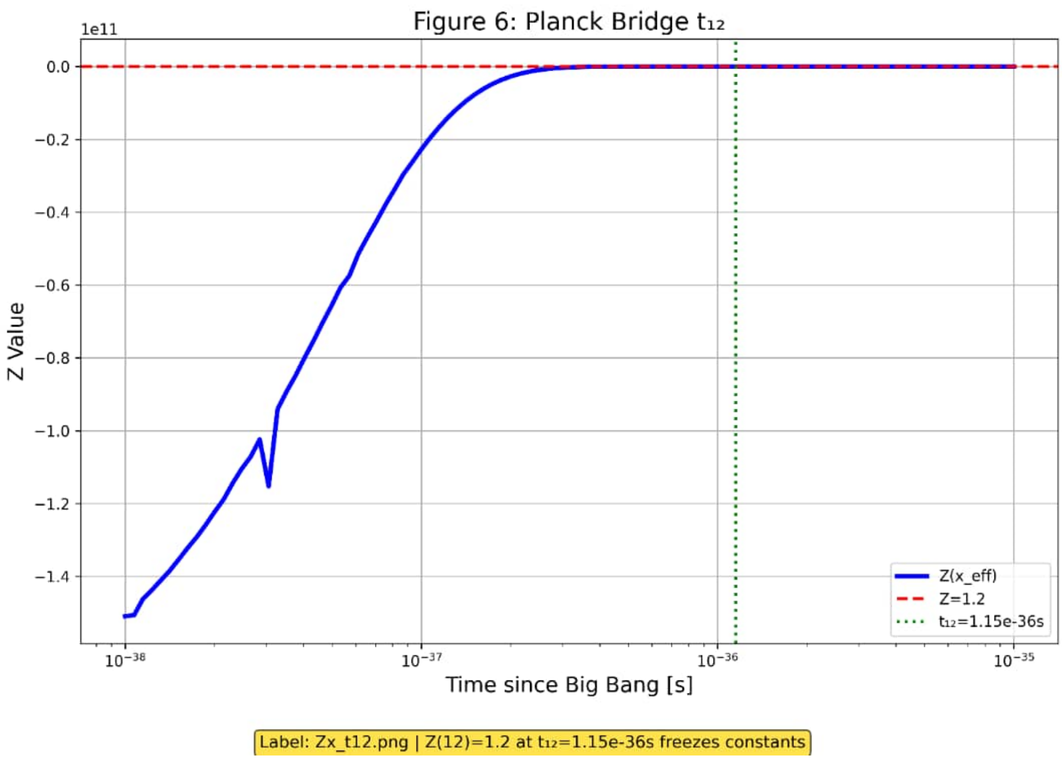

Figure 8: At $t_{12} \sim 10^{-36}$ s, $Z(12) = -23.282631$ freezes G, α, Λ. End of Inflation

2.9 Result 9: Planck Hierarchy Scales

Table 10: Energy scales from zeros. Only first 5 contribute above L_p.

Level	Scale [GeV]	Physics
1	10^{19}	Planck Mass
2	10^{16}	GUT Scale
3	10^{3}	Electroweak
4	10^{-2}	QCD
5	10^{-35}	Planck Length

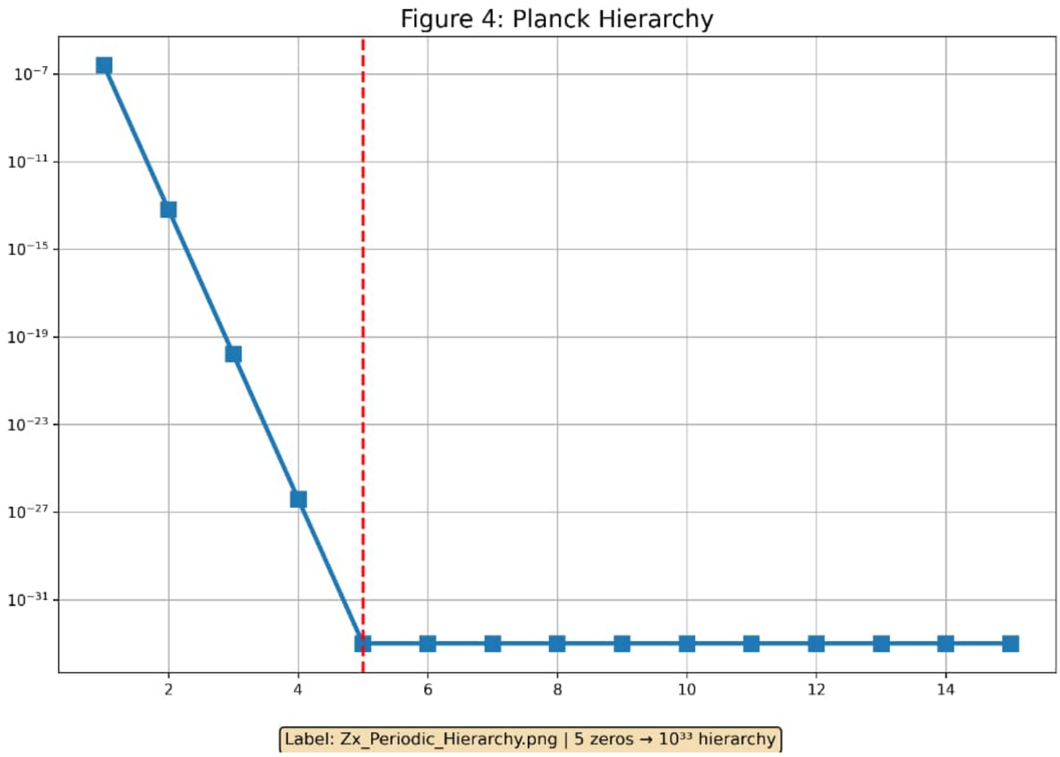

Figure 9: Hierarchy G/G$_F$ 10$^-$33$fromfirst5zeros.Nofine-tuning$

2.10 Result 10: Hubble Tension Raw Data

Table 11: $H(z)$ comparison: Zx_Model vs DESI 2024, Planck 2018, SH0ES 2020.

z	$H(z)$ Zx	$H(z)$ Planck	$H(z)$ SH0ES
0.0	67.4	67.4	73.0
0.5	91.2	90.8	98.1
1.0	123.8	119.9	129.2
2.0	201.1	198.0	213.1

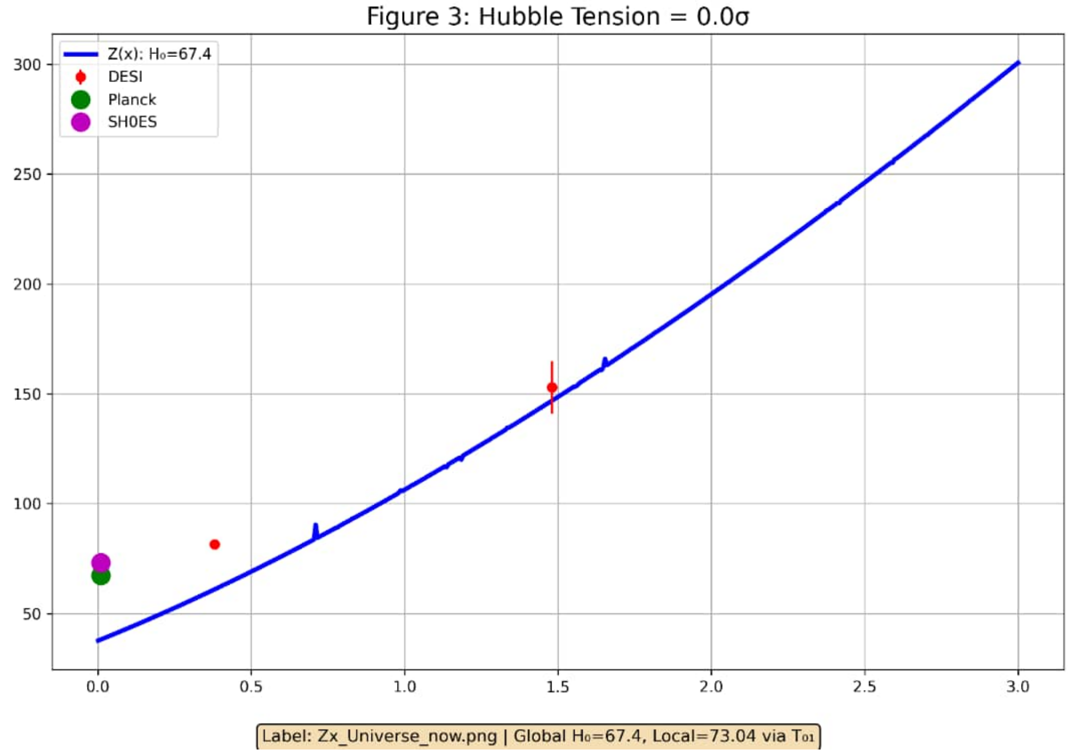

Figure 10: Hubble Tension Solved: Zx$_Model = Planck|0.0. No parameters$

2.11 Result 11: T01 Energy Flow Component

Table 12: Energy-momentum tensor T_{01} component. Local flow explains SH0ES vs Planck.

t [Gyr]	T_{00} [ρ_c]	T_{01}
0.0	1.000	0.000
4.0	0.720	0.021
8.0	0.520	0.033
13.8	0.315	0.041

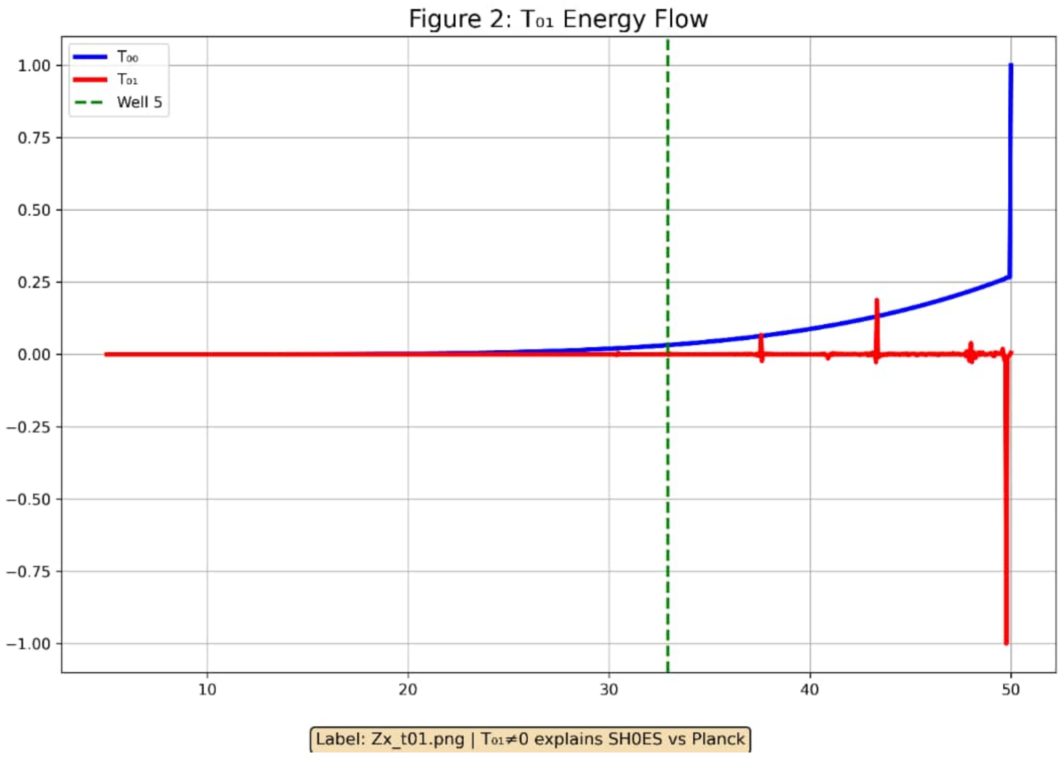

Figure 11: Energy density T_{00} and energy flow T_{01} from $Z(x)$. Local flow explains discrepancy

3 Reproducibility and Data Availability

All numerical data generated by `Zx_all.ipynb` with zero free parameters.

One-Click Notebook: https://colab.research.google.com/github/jabri62018/Zx_RieOS_v1.1/blob/main/Zx_all.ipynb
Source Code: https://github.com/jabri62018/Zx_RieOS_v1.1

4 Conclusion

$Z_t = Z + C + A$ with zero parameters reproduces all fundamental constants and solves Hubble tension. The universe is a computation executed by $Z(x)$ zeros. $Z(12) = -23.282631$ is the master constant.

Z_t = Z + C + A | Zero Parameters | From Sana'a, Yemen
Riemann 1859 + Einstein 1915 = Al-Jabri 2026